

Input/Output

Scanner in = new Scanner(System.in)

↓
Standard Input Stream =

import java.util.Scanner

nextInt()
nextLine()
next()

Console class

To read even invisible one

Console cons = System.console()

cons.readLine()
↓
gets

cons.readPassword()

↓
array of characters



Example

printf → format

S.O.B.L ; ,

multiple flags can be also
be used ,

To read from file



Scanner object

PrintWriter



Create file

→ File
can
create

Scanner sc = new Scanner (path.of () ,
"UTF-8")



Read from file

PrintWriter wt = new PrintWriter
(~~path~~ file name ,
UTF-8)



create file for you



String

Java

String



sequences of Unicode characters

"Java | u 2 | 22 " → S
↑ ↓ ↑
char char char

Java standard library →
String class

" " → empty String

Substrings

Extract string from larger string

"Hello"
↓

substring(0, 3) → Hel

code units / code points → 0

(0, 3) → from 0 to 2 3 (exclusive)

length of substring

$$(i, j) = j - i //$$



Concatenation
operator +

"Hello" + "World"
= "HelloWorld"

Delimiter

String.join("/", "H", "E", "L");

~~HEL~~ H|E|L

java -11 Repeat

"Hi".repeat(3)
HiHiHi

Strings → immutable

You cannot change the characters

String share
same pool

If you abandon object string
s = null

Garbage collector automatically
deletes it

a = "HelloWorld"
b = a

a → HelloWorld
b →



Testing Equality

equals

s } string
t }

s.equals(t)

s == t only checks
whether they are
in same location

Empty String

str.length() == 0

✓
empty string

Null String

str = null; → no object
associated

you cannot use any
method try to use

"NullPointerException"

Java has
Many string methods

1. toLowerCase()

2. isEmpty()

3. toUpperCase()

4. indexOf()

